Project Report
Interferometric Characterization of Silicon-on-Insulator Waveguides Using Mach–Zehnder and Michelson Architectures

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Abstract

This report presents the design, simulation, and experimental characterization of silicon-on-insulator (SOI) waveguide devices aimed at accurately extracting the group index of silicon strip waveguides. Two interferometric architectures were implemented: a Mach–Zehnder interferometer (MZI), consisting of grating couplers, Y-branch splitters, and unbalanced optical paths; and a Michelson interferometer (MI), incorporating grating couplers, 3-dB directional couplers, unbalanced optical paths, and mirror waveguides. In both devices, the wavelength-dependent interference fringes enable precise determination of the group index through the measured free spectral range (FSR). Numerical mode simulations provide the effective and group indices required for analytical transfer-function models, which show excellent agreement with experimental data. Owing to its round-trip phase accumulation, the MI achieves the same FSR as an MZI with twice the physical arm-length mismatch, enabling a more compact footprint. Together, the MZI and MI measurements establish a consistent and validated methodology for extracting dispersion parameters of SOI waveguides.

I. Introduction

Integrated photonics enables compact, low-loss interferometric devices for precise characterization of waveguide properties. Among these, the Mach–Zehnder interferometer (MZI) and the Michelson interferometer (MI) serve as fundamental building blocks for phase-sensitive measurements in silicon photonics [1], [2]. The MZI is widely used for sensing, modulation, and dispersion analysis, while the MI offers a reflective architecture advantageous for compact layouts and efficient reference-path control.

In this work, both interferometer types are designed, modeled, fabricated and experimentally characterized to determine the group and effective indices of silicon strip waveguides on a silicon-on-insulator (SOI) platform. The silicon waveguide height is fixed at 220 nm by the fabrication process, and the width is chosen as 500 nm. The devices operate under quasi-TE polarization, enabled by polarization-matched grating couplers at the input and output [3]. Unbalanced interferometers are designed with arm length differences $\Delta L = L_2 - L_1$ chosen to ensure that the free spectral range (FSR) lies within the measurement bandwidth of approximately 50 nm.

The MZI, an artistic representation of which is shown in Fig. 1, consists of grating couplers, Y-branch splitters, and unbalanced optical paths, whereas the MI incorporates grating couplers, 3-dB couplers, mirror waveguides, and unbalanced paths. Experimental results, including transmission spectra and extracted n_{eff} and n_{g} values, are presented, providing a complete characterization of both interferometer configurations.

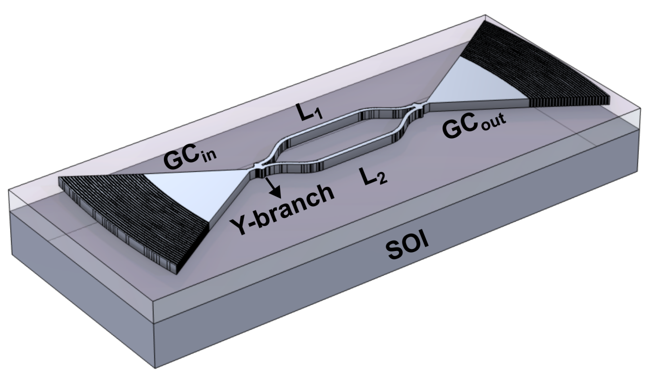

Fig. 1: Artistic depiction of a SOI-based MZI with arm lengths L_1 and L_2, input and output grating couplers ($\mathrm{GC_{in}}$ and $\mathrm{GC_{out}}$), and Y-branch splitters. The cladding is omitted for clarity.

II. Theory

A. Waveguide

The complex wavelength-dependent effective index n_{eff} of a guided mode is defined as:

$$n_{\text{eff}}(\lambda) = \frac{\beta\lambda}{2\pi}, \tag{1}$$

where β is the complex propagation constant and λ is the free-space wavelength.

The group index n_g, which governs pulse propagation and determines the spectral spacing of interference fringes, is related to n_{eff} by:

$$n_g = n_{\text{eff}} - \lambda\frac{dn_{\text{eff}}}{d\lambda}. \tag{2}$$

B. Mach–Zehnder Interferometer

Consider the Mach–Zehnder interferometer (MZI) sketched in Fig. 2. The input field (complex amplitude) is denoted E_i. The input is split into two arms which we label upper (1) and lower (2). The arms have physical lengths L_1 and L_2, effective indices $n_{\text{eff},1}(\lambda)$ and $n_{\text{eff},2}(\lambda)$, and power attenuation coefficients α_1 and α_2. The input splitter is assumed to be ideal and

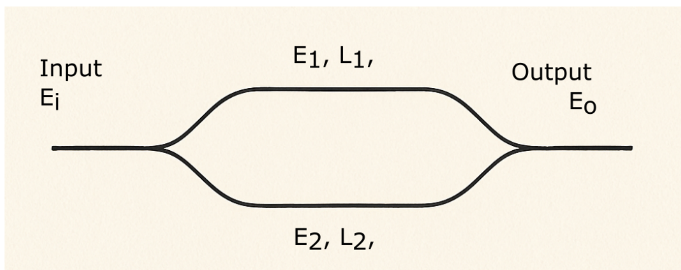

Fig. 2: A simple schematic of an unbalanced MZI with arm length L_1 and L_2.

50:50, so the amplitudes launched into the two arms are

$$E_{1,\text{in}} = \frac{1}{\sqrt{2}}E_i, \qquad E_{2,\text{in}} = \frac{1}{\sqrt{2}}E_i. \tag{3}$$

The propagation constant in arm j is

$$\beta_j(\lambda) = \frac{2\pi}{\lambda}\, n_{\text{eff},j}(\lambda). \tag{4}$$

If power attenuates as $e^{-\alpha_j z}$, then the field accumulates a factor $e^{i\beta_j L_j}e^{-\frac{1}{2}\alpha_j L_j}$. Thus the fields arriving at the second splitter are

$$E_{1,\text{out}} = \frac{1}{\sqrt{2}}E_i\, e^{i\beta_1 L_1}e^{-\frac{1}{2}\alpha_1 L_1}, \qquad E_{2,\text{out}} = \frac{1}{\sqrt{2}}E_i\, e^{i\beta_2 L_2}e^{-\frac{1}{2}\alpha_2 L_2}. \tag{5}$$

At the second 50:50 coupler the fields recombine, giving

$$E_o = \frac{1}{\sqrt{2}}E_{1,\text{out}} + \frac{1}{\sqrt{2}}E_{2,\text{out}} = \frac{E_i}{2}\Big(e^{i\beta_1 L_1}e^{-\frac{1}{2}\alpha_1 L_1} + e^{i\beta_2 L_2}e^{-\frac{1}{2}\alpha_2 L_2}\Big). \tag{6}$$

The transmission, normalized to input power, is

$$T_{\text{MZI}}(\lambda) = \frac{|E_o|^2}{|E_i|^2} \tag{7}$$

$$= \frac{1}{4}\Big(e^{-\alpha_1 L_1} + e^{-\alpha_2 L_2} + 2\, e^{-\frac{1}{2}\alpha_1 L_1}e^{-\frac{1}{2}\alpha_2 L_2}\cos[\Delta\phi_{\text{MZI}}(\lambda)]\Big), \tag{8}$$

where the phase difference is

$$\Delta\phi_{\text{MZI}}(\lambda) = \beta_2(\lambda)L_2 - \beta_1(\lambda)L_1 = \frac{2\pi}{\lambda}\big(n_{\text{eff},2}(\lambda)L_2 - n_{\text{eff},1}(\lambda)L_1\big). \tag{9}$$

If both arms have the same loss per unit length ($\alpha_1 = \alpha_2 = \alpha$) and the effective indices are equal ($n_{\text{eff},1} = n_{\text{eff},2} = n_{\text{eff}}$), then the transmission reduces to

$$T_{\text{MZI}}(\lambda) \approx \frac{1}{2}\big[1 + \cos(\Delta\phi(\lambda))\big], \tag{10}$$

with

$$\Delta\phi_{\text{MZI}}(\lambda) = \frac{2\pi}{\lambda}\, n_{\text{eff}}(\lambda)\,\Delta L, \qquad \Delta L = L_2 - L_1. \tag{11}$$

The free spectral range (FSR) is the wavelength spacing between adjacent maxima (or minima). It is defined by the condition that the phase difference changes by 2π:

$$\Delta\phi_{\mathrm{MZI}}(\lambda + \mathrm{FSR}_{\mathrm{MZI}}) - \Delta\phi_{MZI}(\lambda) = 2\pi. \tag{12}$$

Linearizing,

$$\left.\frac{d\Delta\phi_{\mathrm{MZI}}}{d\lambda}\right|_{\lambda} \cdot \mathrm{FSR}_{\mathrm{MZI}} \approx 2\pi. \tag{13}$$

Since

$$\Delta\phi_{\mathrm{MZI}}(\lambda) = \frac{2\pi}{\lambda} n_{\mathrm{eff}}(\lambda)\Delta L, \tag{14}$$

its derivative is

$$\frac{d\Delta\phi_{\mathrm{MZI}}}{d\lambda} = -\frac{2\pi}{\lambda^2}\, n_g(\lambda)\,\Delta L, \tag{15}$$

where the group index is defined as

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda}. \tag{16}$$

Thus,

$$\mathrm{FSR}_{\mathrm{MZI}} = \frac{\lambda^2}{n_g(\lambda)\,\Delta L}. \tag{17}$$

From a measured FSR one extracts the group index:

$$n_g(\lambda) = \frac{\lambda^2}{\mathrm{FSR}_{\mathrm{MZI}} \cdot \Delta L}. \tag{18}$$

C. Michelson Interferometer (MI)

A Michelson interferometer (MI) is another fundamental integrated photonic structure used for phase and dispersion measurements. In contrast to the Mach–Zehnder interferometer (MZI), where the optical field traverses two distinct paths and recombines at a second coupler, the MI splits light into two arms and reflects it back toward the input coupler using reflectors (typically terminating mirrors, bragg reflector or loop-back waveguides). The returning fields interfere at the same coupler, producing a spectral interference pattern.

Let the input field be E_i, and assume an ideal 50:50 splitter. The forward and reflected field contributions from arms of length L_1 and L_2 can be written as

$$E_o = \frac{E_i}{2}\left(e^{2i\beta L_1} + e^{2i\beta L_2}\right), \tag{19}$$

where the factor of 2 in the phase exponent accounts for the round-trip propagation.

The normalized transmission (or reflection) is then

$$T_{\mathrm{MI}}(\lambda) = \frac{1}{2}\left[1 + \cos(\Delta\phi_{\mathrm{MI}}(\lambda))\right], \tag{20}$$

with the phase difference

$$\Delta\phi_{\mathrm{MI}}(\lambda) = \frac{4\pi}{\lambda} n_{\mathrm{eff}}(\lambda)\Delta L, \qquad \Delta L = L_2 - L_1. \tag{21}$$

Because the phase accumulates twice for each arm, the free spectral range (FSR) of the MI is

$$\mathrm{FSR}_{\mathrm{MI}} = \frac{\lambda^2}{2\, n_g(\lambda)\,\Delta L}. \tag{22}$$

The key difference between MI and MZI operation is that in the MI light experiences a *round-trip phase*, effectively doubling the phase sensitivity:

- In MZI: $\Delta\phi_{\mathrm{MZI}} = \frac{2\pi}{\lambda} n_{\mathrm{eff}}\Delta L$
- In MI: $\Delta\phi_{\mathrm{MI}} = \frac{4\pi}{\lambda} n_{\mathrm{eff}}\Delta L$

Thus, to achieve the same FSR,

$$\Delta L_{\mathrm{MI}} = \frac{1}{2}\Delta L_{\mathrm{MZI}}. \tag{23}$$

This property makes the MI attractive for compact phase-sensitive devices and for applications where enhanced phase sensitivity or footprint reduction is desired.

III. Modeling and Simulations

Lumerical MODE is used to model the cross-section of the strip waveguide and to calculate dispersion parameters. Lumerical INTERCONNECT is employed to simulate the complete Mach–Zehnder interferometer (MZI) circuit using SiePIC library components. MATLAB is used to theoretically model the MZI transfer function and for data analysis and plotting. For the layout design, Luceda IPKISS will be used.

A. Strip Waveguide

We use a silicon strip waveguide with thickness *H* = 220 nm and width *W* = 500 nm, surrounded by silicon dioxide (SiO_2) as both substrate and cladding. The refractive index as a function of wavelength λ was taken from Palik's handbook [4], which is also available in the material library of Lumerical. The simulation domain was set to 4 μm × 4 μm in the x- and y-dimensions, respectively, with metallic boundary conditions applied at each boundary.

The simulated cross-sectional (xy-plane) intensity mode profiles for quasi-TE and quasi-TM polarizations at $\lambda = 1500$ nm are shown in Fig. 3. The effective mode indices were found to be $n_{\text{eff}} = 2.501$ for the TE mode and $n_{\text{eff}} = 1.8391$ for the TM mode. The corresponding propagation losses were calculated as 4.71×10^{-4} dB/cm for the TE mode and 3.5×10^{-4} dB/cm for the TM mode. These losses were taken into account in the theoretical modeling of the MZI transfer function. In the remainder of this report, the focus is on a TE-mode-based MZI circuit, as it is the intended mode to be excited in this project.

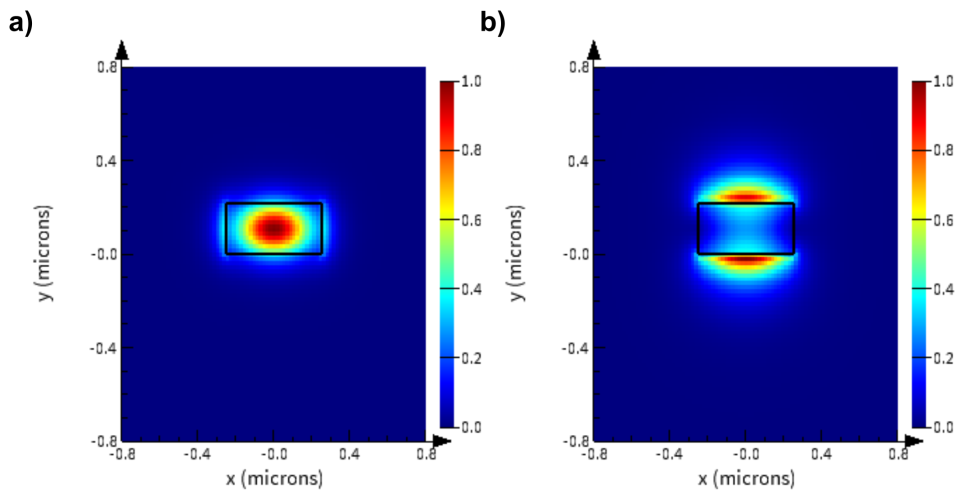

Fig. 3: Simulated fundamental modes of the 500 nm wide, 220 nm thick silicon strip waveguide on SOI at λ = 1500 nm: (a) TE mode and (b) TM mode.

In addition to modal analysis at $\lambda = 1500$ nm, a wavelength sweep was performed to obtain the dispersion of the TE mode. Figure 4a shows the dependence of n_{eff} on λ. The wavelength dependence of the effective index can be approximated by a truncated Taylor-series expansion around a central wavelength λ_0, typically chosen as 1550 nm:

$$n_{\text{eff}}(\lambda) \approx n_0 + \left.\frac{dn_{\text{eff}}}{d\lambda}\right|_{\lambda_0} (\lambda - \lambda_0) + \frac{1}{2}\left.\frac{d^2 n_{\text{eff}}}{d\lambda^2}\right|_{\lambda_0} (\lambda - \lambda_0)^2, \tag{24}$$

where the first derivative term accounts for the group index, and the second derivative term represents the group velocity dispersion (GVD).

The group index n_g is related to the effective index by:

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{25}$$

Alternatively, the same dependence can be expressed in polynomial form as:

$$n_{\text{eff}}(\lambda) \approx n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2, \tag{26}$$

where n_0, n_1, and n_2 are fitting coefficients. The fitting was performed in MATLAB using the simulated dispersion data, and the extracted coefficients, given below, were then used for subsequent modeling of the interferometer transfer function.

$$n_{\text{eff}}(\lambda) \approx 2.4444 - 1.1315(\lambda - \lambda_0) - 0.0395(\lambda - \lambda_0)^2, \tag{27}$$

Moreover, the group index n_g was also calculated from the effective index using Eq. 25. Figure 4b shows the group index, which will be compared with the experimentally extracted group index.

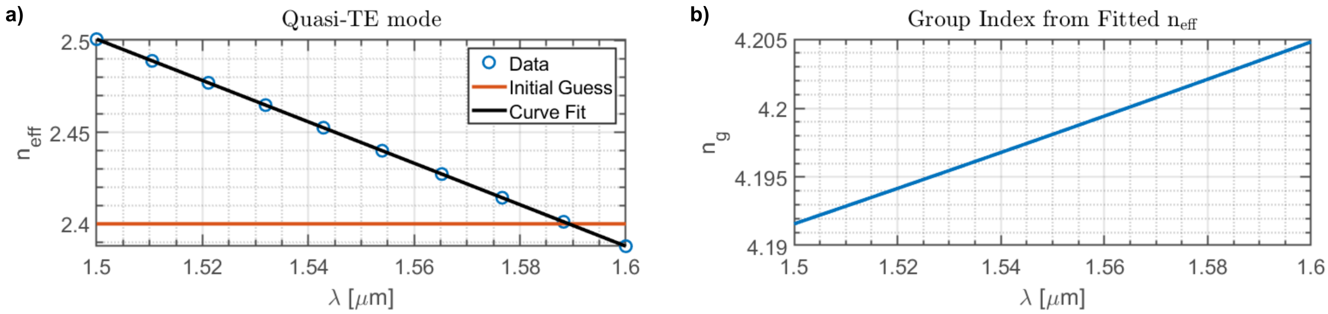

Fig. 4: Dispersion analysis of the strip waveguide: (a) Effective index (n_{eff}) as a function of wavelength (λ). Blue circles represent simulated data, while the black solid line is the fit. (b) Group index (n_g) as a function of wavelength (λ), calculated from n_{eff}.

B. MZI and MI

The MZI and MI were implemented in Lumerical INTERCONNECT, and the corresponding transfer functions (Eq. 10 and Eq. 20) were simulated in MATLAB. The grating coupler and Y-branches were imported from the design kit provided with this course. The responses of the grating coupler and Y-branch, in terms of gain (dB), are shown in Fig. 5a and 5b, respectively.

The circuit schematics for the MZI is shown in Fig. 6. The MZI consists of Y-branches "SPAR_1" and "SPAR_2" as splitter and combiner, respectively. The grating couplers are labeled as "SPAR_3" and "SPAR_4" (input and output). The waveguides are labeled as "WG_1" and "WG_2", where "WG_1" is kept fixed with length L. To introduce unbalance, the length of "WG_2" is varied as $L + \Delta L$. Interference at the output is thus observed by varying ΔL. The optical network analyzer "ONA_1" is used to determine the circuit response. Below is the list of ΔL values for the different MZI layouts designed.

In addition, the circuit schematic of the MI is shown in Fig. 7. The MI circuit consists of input and output grating couplers labeled "SPAR_1" and "SPAR_2," respectively. Both grating couplers are connected to a bi-directional coupler (BDC1), labeled "ebeam_bdc_te1550_1." The BDCs are imported from the design kit provided with this course. BDC1 is further connected to another bi-directional coupler (BDC2, "ebeam_bdc_te1550_2") through two waveguides, labeled "WGD_1" and "WGD_2," with lengths L and $L + \Delta L$, respectively. Finally, the outputs of BDC2 are looped back via another waveguide, "WGD_3," which functions as a mirror.

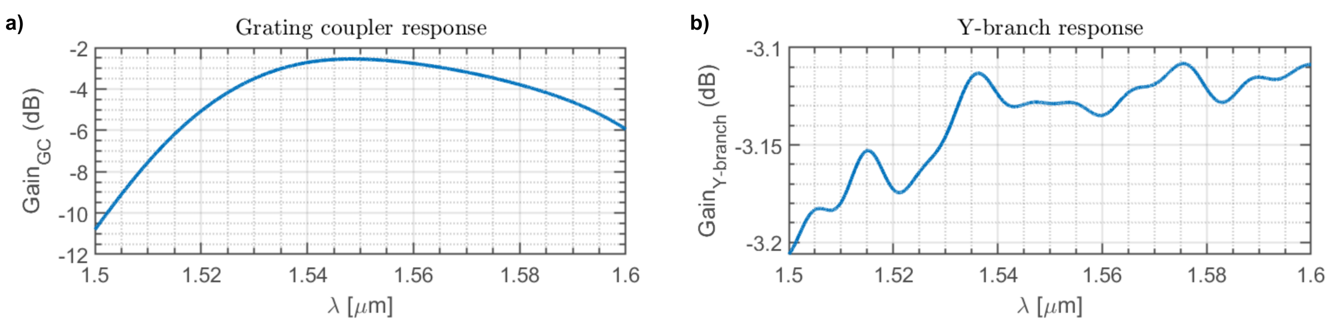

Fig. 5: Gain as a function of wavelength: (a) Grating coupler. (b) Y-branch.

Figure 8 presents sample gain (dB) plots for (a) an MZI with $\Delta L = 50$ μm and (b) an MI with $\Delta L = 25$ μm. The results compare the theoretically calculated transfer function (blue solid line) with the Lumerical INTERCONNECT simulation (red solid line). The comparison demonstrates that the MI with half the path length difference ($\Delta L/2$) exhibits an identical spectral response to the MZI, both in terms of fringe spacing and overall interference pattern. The two results show good agreement in the positions of the maxima and minima. The slight discrepancy in gain values arises because the theoretical model neglects insertion losses from the grating couplers and Y-branches, whereas the simulation includes these effects (see Fig. 5). The theoretical model, however, accounts for the propagation loss of the waveguides extracted from Lumerical mode simulations.

Furthermore, the free spectral range (FSR) as a function of wavelength for both interferometers was calculated and simulated, with the comparison shown in Fig. 9. Consistent with the gain results, the MI with $\Delta L = 25$ μm exhibits the same FSR as the MZI with $\Delta L = 50$ μm, confirming that the MI effectively doubles the optical path difference through reflection, thereby reproducing the same interference behavior. The small mismatch between theoretical and simulated results can again be attributed to the exclusion of grating coupler insertion losses in the theoretical model.

Finally, an FSR analysis was carried out for a range of length mismatches ΔL at a fixed wavelength of $\lambda = 1.55$ μm. Figure 10(a) shows the FSR for the MZI as a function of ΔL, ranging from 50 µm to 150 µm, while Fig. 10(b) presents the corresponding results for the MI, with ΔL varying from 25 µm to 75 µm. As expected, the FSR decreases with increasing

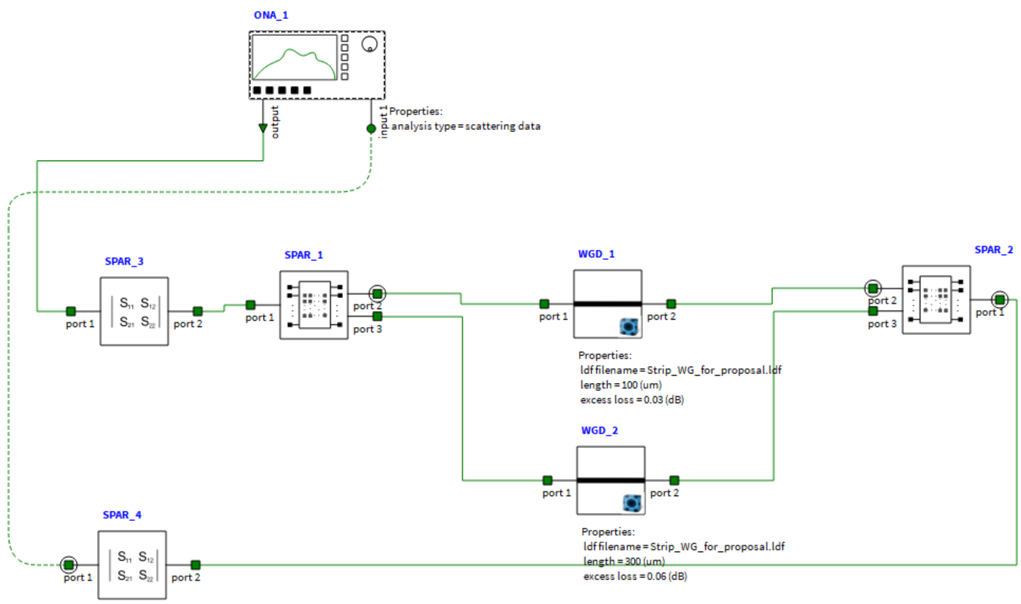

Fig. 6: Schematic of the MZI implemented in Lumerical INTERCONNECT.

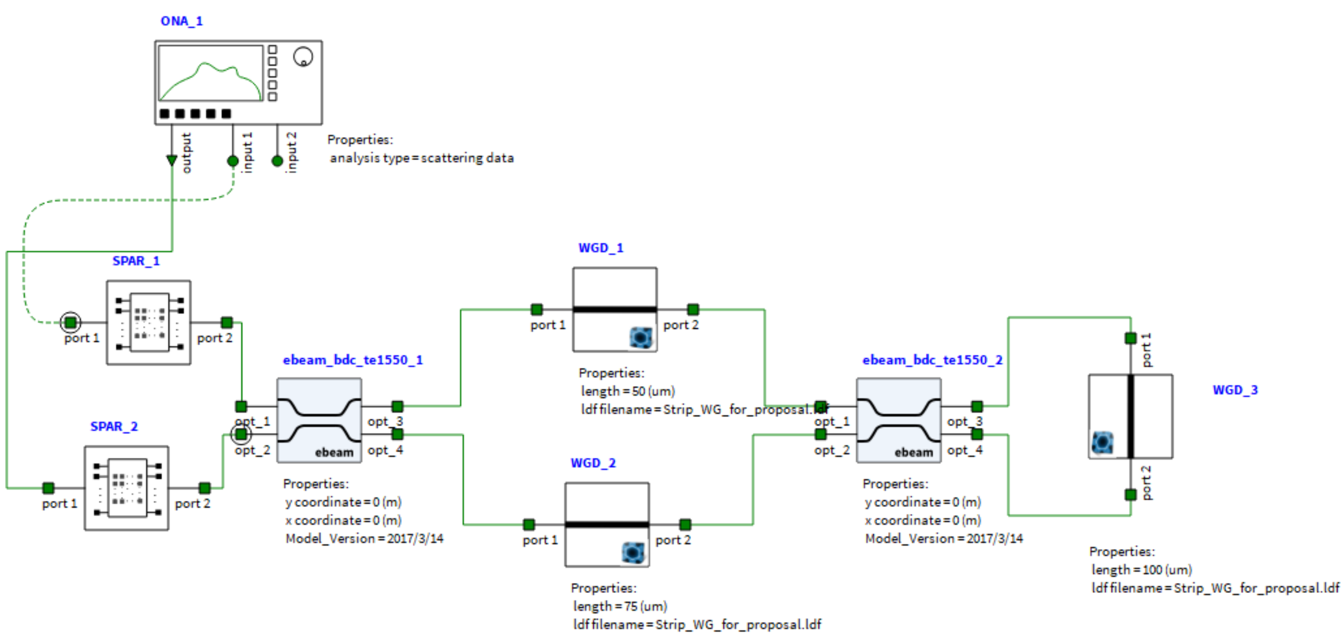

Fig. 7: Schematic of the MI implemented in Lumerical INTERCONNECT.

ΔL, since a larger path length difference enhances the phase sensitivity of the interferometer, resulting in a smaller spectral spacing between adjacent transmission maxima. Moreover, the results confirm that the MI exhibits the same FSR behavior as the MZI when its path length difference is half that of the MZI, consistent with the theoretical expectation that the MI effectively doubles the optical path difference through reflection.

By designing interferometers with different unbalance lengths, the FSR can be tuned to fall within the measurable wavelength range, enabling more accurate extraction of the group index. The corresponding values are summarized in Table I, and these ΔL values were targeted for fabrication.

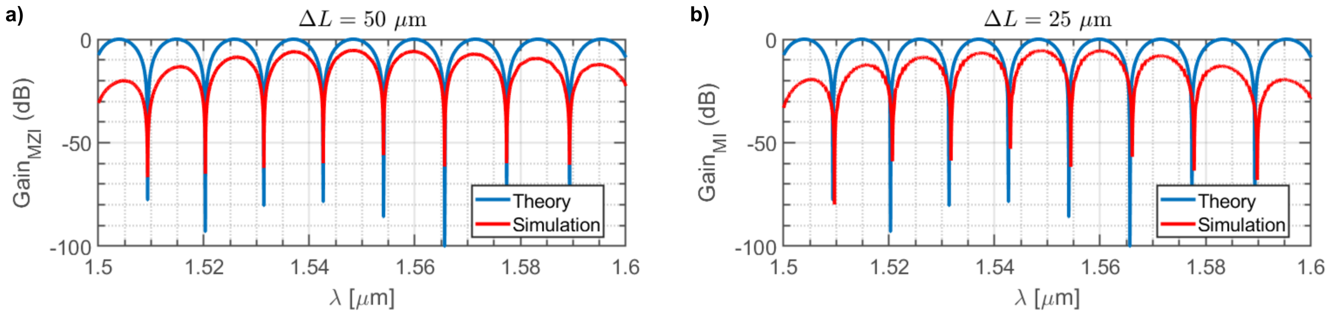

Fig. 8: Interference spectrum (gain) of an (a) MZI with $\Delta L = 50\ \mu$m and (b) MI with $\Delta L = 25\ \mu$m. The blue solid line corresponds to the theoretical transfer function, while the red line shows the Lumerical INTERCONNECT simulation result.

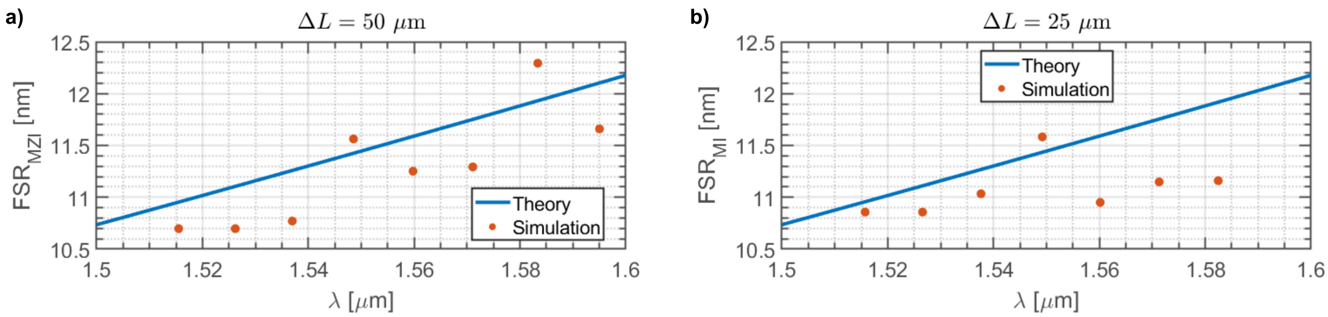

Fig. 9: FSR of an (a) MZI with $\Delta L = 50\ \mu$m and (b) MI with $\Delta L = 25\ \mu$m. The blue solid line corresponds to the theoretical calculation, while the red dots shows the Lumerical INTERCONNECT simulation result.

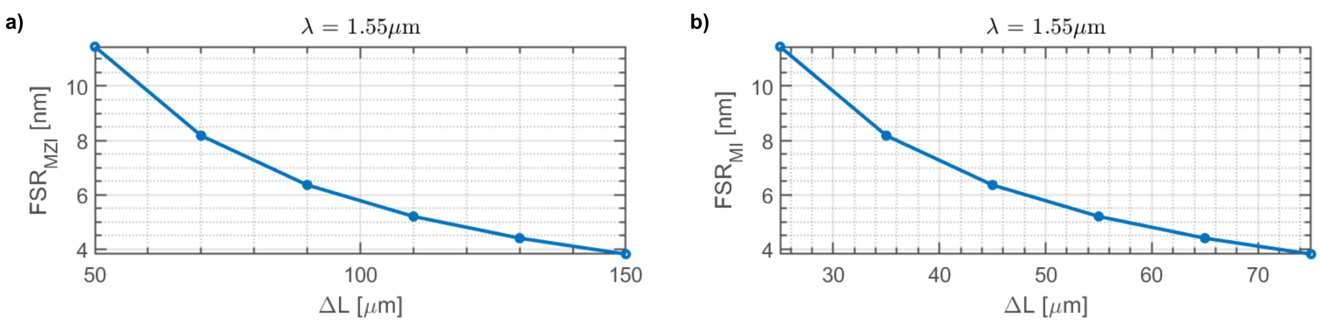

Fig. 10: FSR of the (a) MZI and (b) MI as a function of waveguide length mismatch ΔL at wavelength λ = 1.55 μm.

TABLE I: FSR values extracted from simulation for different ΔL at λ = 1.55 μm.

ΔL_{MZI} (μm)	ΔL_{MI} (μm)	FSR (nm)
50	25	11.4488
70		8.1777
90		6.3605
110	55	5.2040
145		3.9478
150	75	3.8163

IV. Fabrication

A. Fabrication Variability and Corner Analysis

In integrated photonic circuits, the optical response of waveguide-based devices is highly sensitive to geometrical variations introduced during fabrication. Small deviations in waveguide width, height, or etch depth can lead to measurable shifts in the effective and group indices, thereby altering the designed interference conditions and overall device performance. Such variations commonly arise from a combination of factors including e-beam or photolithography resolution limits, resist development nonuniformities, etch rate fluctuations, and deposition thickness variations during film growth. Surface roughness and sidewall angle deviations further contribute to additional propagation loss and phase errors, especially in high-index-contrast waveguides such as those on the silicon-on-insulator (SOI) platform.

To assess the impact of these fabrication tolerances on the performance of the designed interferometric circuits, a **corner**

analysis was carried out. In this analysis, nine simulations were performed by systematically varying the core waveguide width (W) and height (H) around their nominal values. The nominal dimensions were set to $W = 0.50$ μm and $H = 0.22$ μm, corresponding to the target design parameters. The width was allowed to vary within ± 0.03 μm (i.e., from 0.47 μm to 0.51 μm), while the height was varied within ± 0.005 μm (i.e., from 0.215 μm to 0.223 μm).

This results in a 3×3 matrix of corner cases, representing combinations of minimum, nominal, and maximum dimensions for both width and height:

$$(W, H) = \begin{bmatrix} (0.47, 0.215) & (0.47, 0.220) & (0.47, 0.223) \\ (0.50, 0.215) & (0.50, 0.220) & (0.50, 0.223) \\ (0.51, 0.215) & (0.51, 0.220) & (0.51, 0.223) \end{bmatrix} \ [\mu\text{m}]$$

For each corner combination, the effective index (n_{eff}), group index (n_g), and propagation loss were extracted from mode simulations. The results were subsequently used to evaluate the performance variation in both the Mach–Zehnder and Michelson interferometers. This analysis enables identification of the sensitivity of the interferometers to fabrication tolerances and provides insight into the expected device-to-device variability in the fabricated chips.

Figure 11 shows the variation of the effective refractive index (n_{eff}) with wavelength for the different corner cases. A noticeable shift in n_{eff} can be observed with increasing waveguide width and height, which directly affects the optical path length difference in the MZI and MI arms. Similarly, Figure 12 presents the group index (n_g) dependence on wavelength, where small but systematic variations are seen across the corners.

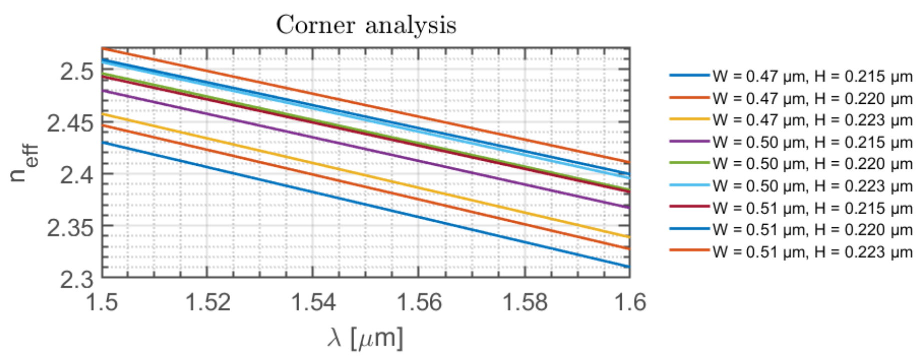

Fig. 11: Effective refractive index (n_{eff}) variation as a function of wavelength for different fabrication corners.

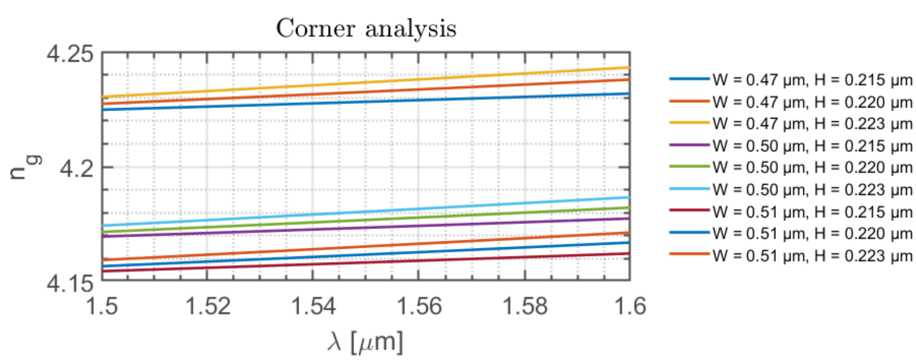

Fig. 12: Group index (n_g) variation as a function of wavelength for different fabrication corners.

The transmission spectra of the MZI, as an example, with a path length difference of $\Delta L = 50$ μm are shown in Figure 13. The results demonstrate that fabrication-induced dimensional deviations lead to noticeable shifts in the spectral response, including changes in the free spectral range (FSR) and extinction ratio. This analysis highlights the importance of robust design practices and fabrication-aware simulations to ensure consistent device performance under realistic process variations. To further evaluate the sensitivity of the MZI design to fabrication-induced waveguide variations, we also plot the Free Spectral Range (FSR) as a function of λ for all corner geometries with $\Delta L = 50$ μm in Fig. 14. At $\lambda = 1.55$ μm, the maximum deviation from the nominal FSR is $-1.5\,\%$. This value indicate that the impact of waveguide dimensional variations on the FSR is very small, confirming that the MZI design is highly robust, in terms of the FSR, against typical fabrication tolerances.

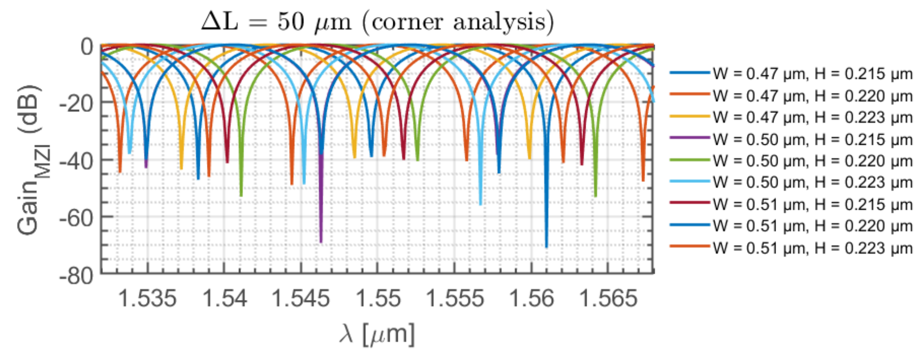

Fig. 13: Simulated MZI transmission spectra ($\Delta L = 50\ \mu$m) under different fabrication corners showing wavelength shift and extinction ratio variation.

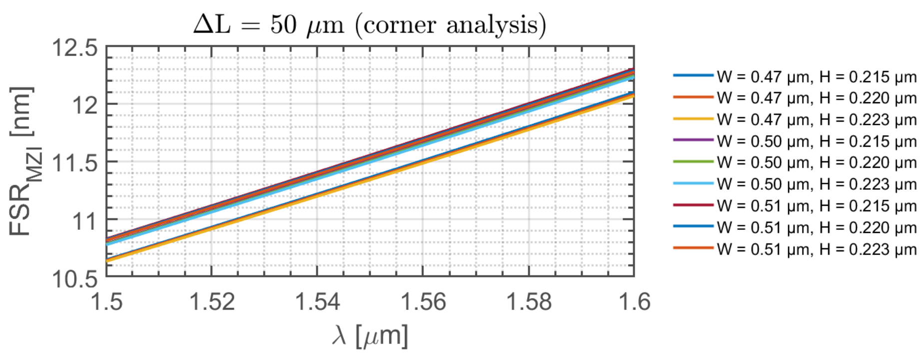

Fig. 14: Simulated MZI FSR ($\Delta L = 50\ \mu$m) under different fabrication corners showing very little deviation from nominal waveguide geometry.

B. Chip layout

The final chip layout is shown in Fig. 15, comprising a set of Mach–Zehnder Interferometers (MZIs) with path-length differences of $\Delta L = 0, 50, 70, 90, 110, 145,$ and $150\ \mu$m. Two de-embedding structures (calibration loop-back) were included to enable calibration and remove the contributions of the grating couplers and waveguide bends. Note that the MZIs with $\Delta L = 0\ \mu$m also serves as calibration structures to remove the contribution of Y-branches. In addition, three MZIs with $\Delta L = 0, 50,$ and $70\ \mu$m were replicated to assess device-to-device reproducibility.

Three Michelson Interferometers (MIs) with $\Delta L = 25, 52.5,$ and $75\ \mu$m were also incorporated into the layout. Table II summarizes all fabricated devices together with their cell names in the draft GDS file and the corresponding path-length differences ΔL.

TABLE II: List of fabricated devices with their cell names and corresponding ΔL.

Cell name	**Device**	$\Delta L[\mu m]$
REF1	De-embed	n/a
REF2	De-embed	n/a
MZI1	MZI	0
REPMZI1	MZI	0
MZI2	MZI	50
REPMZI2	MZI	50
MZI3	MZI	70
REPMZI3	MZI	70
MZI4	MZI	90
MZI5	MZI	110
MZI6	MZI	145
MZI7	MZI	150
MI1	MI	25
MI2	MI	52.5
MI3	MI	75

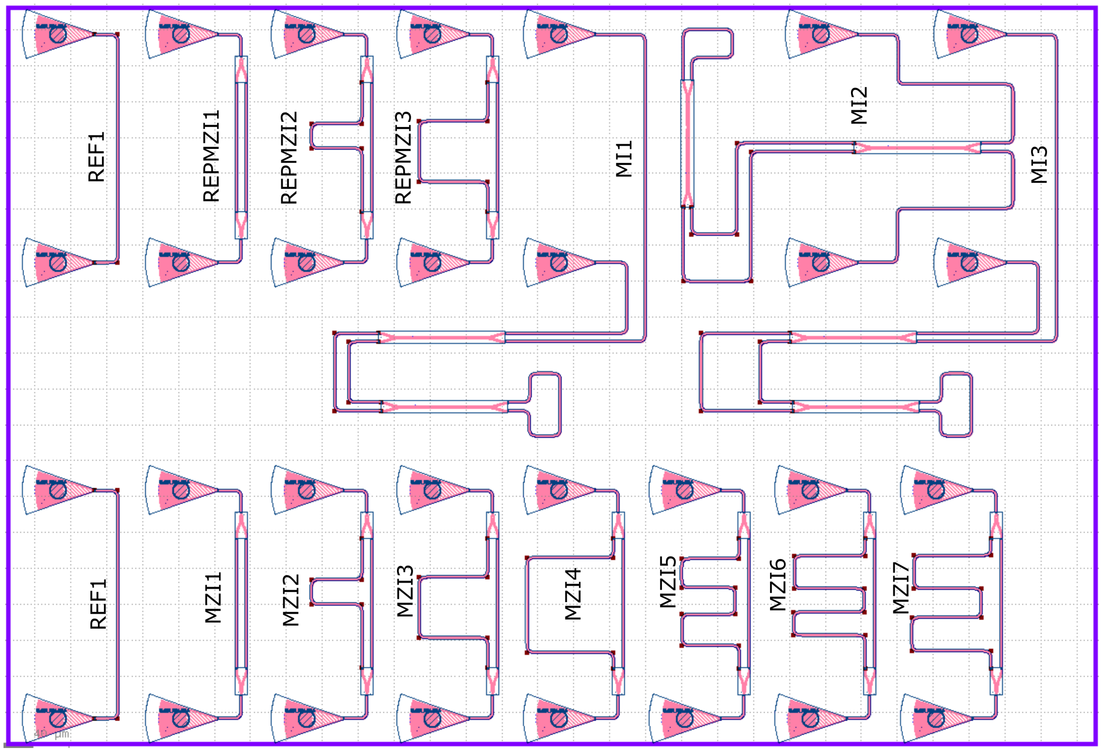

Fig. 15: Final chip layout showing Mach–Zehnder Interferometers (MZIs) with path-length differences from 0 to 150 μm, de-embedding structures for calibration (REF1 and REF2), replicated MZIs for reproducibility (REPMZI1, REPMZI2, REPMZI3) and three Michelson Interferometers (MIs) with $\Delta L = 25, 55$, and 75 μm. See Table II for device details and GDS cell names.

C. Fabrication process flow

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery. The whole process flow is summarized in Fig. 16.

V. Experimental setup

To characterize the devices, a custom-built automated test setup [1], [5] with automated control software written in Python was used [6]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [3]. A 90º rotation was used to inject light into the TM grating couplers [3]. A polarization maintaining fibre array was used to couple light in/out of the chip [7].

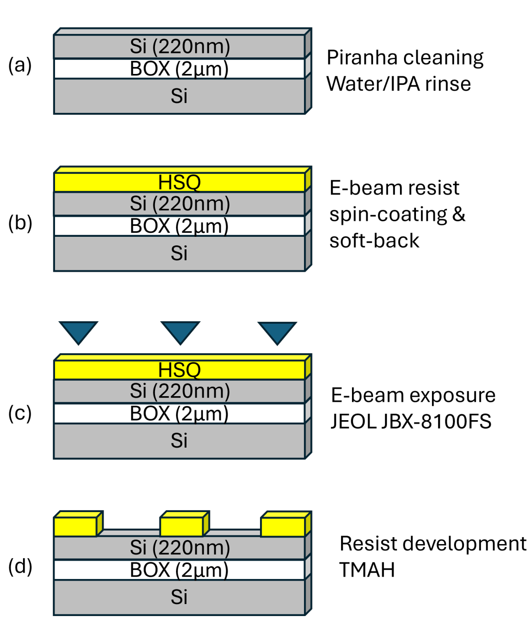

Fig. 16: Fabrication process flow for silicon strip waveguides on a silicon-on-insulator (SOI) platform. (a) Starting substrate consisting of a 220 nm silicon device layer on a 2 µm buried oxide (BOX). (b) Spin coating of HSQ resist followed by a soft-bake step. (c) Electron-beam lithography exposure of the waveguide pattern. (d) Development of the HSQ resist to define the hard-mask pattern. (e) Pattern transfer into the silicon device layer using anisotropic ICP-RIE etching. (f) Removal of the remaining HSQ mask. (g) Deposition of a SiO_2 cladding layer, completing the waveguide fabrication.

VI. Experimental data and analysis

To illustrate the performance of the fabricated devices, we first present the raw experimental data in Figure 17. The data includes the de-embedding structures (Figure 17(a)), Mach–Zehnder Interferometers (MZIs) with $\Delta L = 0$ and 50 μm (Figure 17(b) and (c), respectively), and a Michelson Interferometer (MI) with $\Delta L = 25$ μm (Figure 17(d)). Replicated devices are overlaid in red, demonstrating excellent on-chip device-to-device reproducibility. The de-embedding structures and balanced MZIs serve as calibration references to remove the contributions of grating couplers, Y-branches, and waveguide bends, providing a reliable baseline for further analysis.

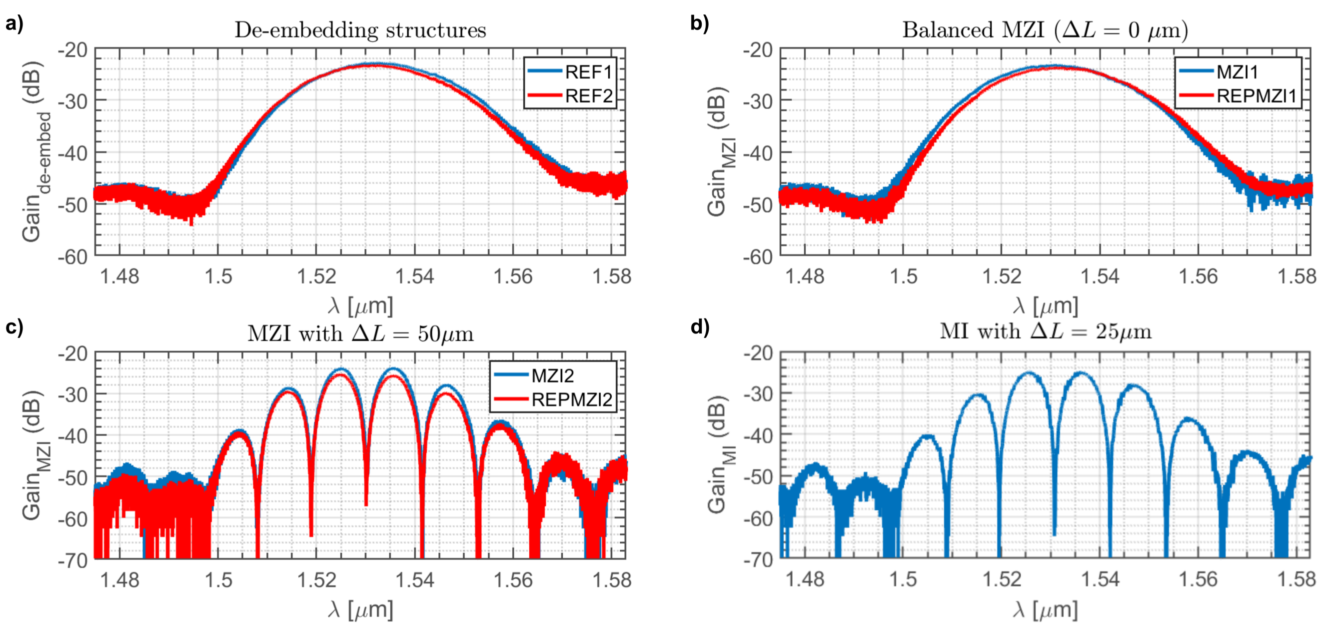

Fig. 17: Raw experimental data for the fabricated devices. (a) De-embedding structures, (b) MZI with $\Delta L = 0$ μm, (c) MZI with $\Delta L = 50$ μm, and (d) MI with $\Delta L = 25$ μm. Replicated devices are shown in red.

Subsequently, the de-embedding (loop-back) structures were fitted using a least-squares method over the -10 dB bandwidth range, and the resulting fit was subtracted from the raw experimental data of the MZIs and MIs. A vertical offset was then applied to baseline the data at 0 dB, enabling direct comparison with the simulation results. As a representative example, Figure 18 shows the calibrated results for (a) an MZI with $\Delta L = 50\ \mu$m and (b) an MI with $\Delta L = 25\ \mu$m. The experimental results (blue lines) are compared with theoretical predictions (red lines) for the nominal waveguide dimensions, $W = 500$ nm and $H = 220$ nm. A slight blue shift is observed in the experimental data, which is consistent with fabrication variability (see Section IV(A)) and remains within the range predicted by the corner analysis (Figure 13). A close inspection of Figure 13 indicates that the experimental results correspond to a waveguide with dimensions $W = 510\ \mathrm{nm}$ and $H = 215\ \mathrm{nm}$. Furthermore, comparison of the experimental spectra clearly shows that a Michelson interferometer achieves the same free spectral range (FSR) as a Mach–Zehnder interferometer when its arm-length mismatch is set to half the value used in the MZI. This directly reflects the doubled phase sensitivity of the MI configuration, which effectively doubles the optical path imbalance for a given physical length difference.

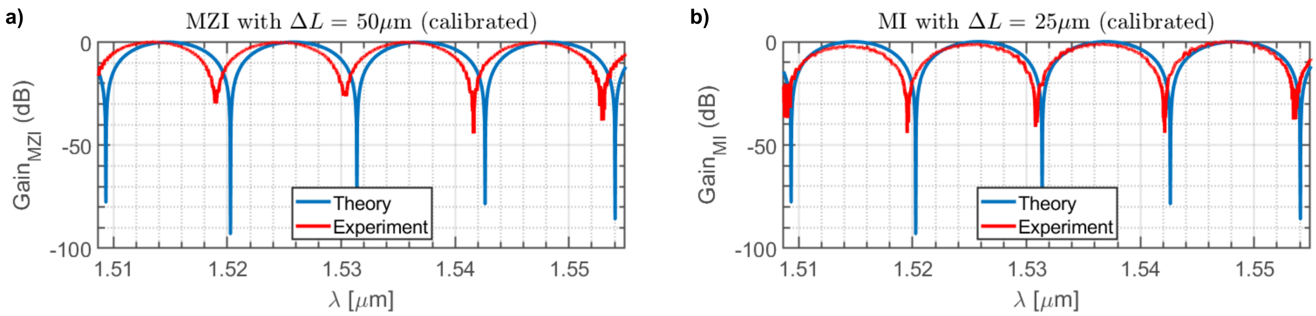

Fig. 18: Calibrated experimental gain (blue) and theoretical prediction (red) for (a) MZI with $\Delta L = 50\ \mu$m and (b) MI with $\Delta L = 25\ \mu$m. Calibration used the loop-back structures fitted over the -10 dB bandwidth and baseline correction to 0 dB.

The effective index (n_{eff}) and group index (n_g) of the interferometer waveguides were extracted from the measured transmission spectra using a peak-finding approach. First, a reference loopback (REF1 in this case) spectrum was used to calibrate the measured data, removing slow background variations by fitting and subtracting a low-order polynomial. The calibrated spectrum was then smoothed using a moving average filter to suppress high-frequency noise.

Transmission minima (fringes) were identified by applying a peak-finding algorithm to the inverted, smoothed spectrum. The positions of these minima, λ_{min}, were used to calculate the free spectral range (FSR) as

$$\mathrm{FSR} = \lambda_{\mathrm{min}}^{(i+1)} - \lambda_{\mathrm{min}}^{(i)}. \tag{28}$$

The group index was then obtained from

$$n_g = \frac{\lambda^2}{\Delta L\, \mathrm{FSR}}, \tag{29}$$

where ΔL is the path length difference of the interferometer.

The effective index n_{eff} was estimated by fitting the measured transmission spectrum with a polynomial model

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \tag{30}$$

where the coefficients n_1, n_2, n_3 were obtained by minimizing the difference between the modeled and measured interferometer response. From $n_{\mathrm{eff}}(\lambda)$, the group index was also independently computed as

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda}. \tag{31}$$

This approach provides simultaneous, consistent extraction of n_{eff} and n_g, accounting for background variations, noise, and dispersion, and is applicable to both MZI and MI interferometers.

In Fig. 19, the extracted (a) $\mathrm{n_{eff}}$ and (b) $\mathrm{n_g}$ for all fabricated MZIs are presented. The experimentally retrieved values are compared against both the simulated nominal waveguide design (W = 200 nm, H = 220 nm) and the full corner analysis. For clarity and conciseness, the results of the corner analysis are shown as a shaded gray region, representing the nine geometrical corner cases considered. The measured $\mathrm{n_{eff}}$ and $\mathrm{n_g}$ exhibit excellent agreement with the nominal simulations (i.e., ≈1% maximum deviation) and fall well within the bounds of the corner-analysis region, confirming the robustness and accuracy of both the extraction method and the fabrication process.

Finally, Fig. 20 presents the extracted (a) $\mathrm{n_{eff}}$ and (b) $\mathrm{n_g}$ for all fabricated MIs. Both the effective index and the group index remain in excellent agreement with the simulated corner-analysis envelope, with a maximum deviation of only ≈1.5% from the nominal geometry. This small discrepancy further confirms the robustness of the MI design against fabrication-induced variations.

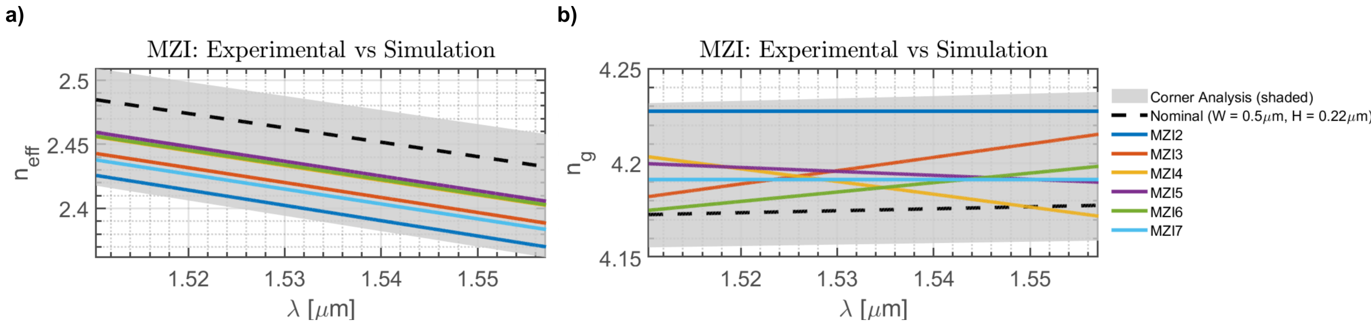

Fig. 19: Extracted (a) $\mathrm{n_{eff}}$ and (b) $\mathrm{n_g}$ from MZIs. Colored curves show experimental results for MZI2 – MZI7 with $\Delta L = 50 - 150$ μm. The black dashed line corresponds to the nominal waveguide geometry (W = 500 nm, H = 220 nm), while the shaded gray region represents the corner-analysis bounds.

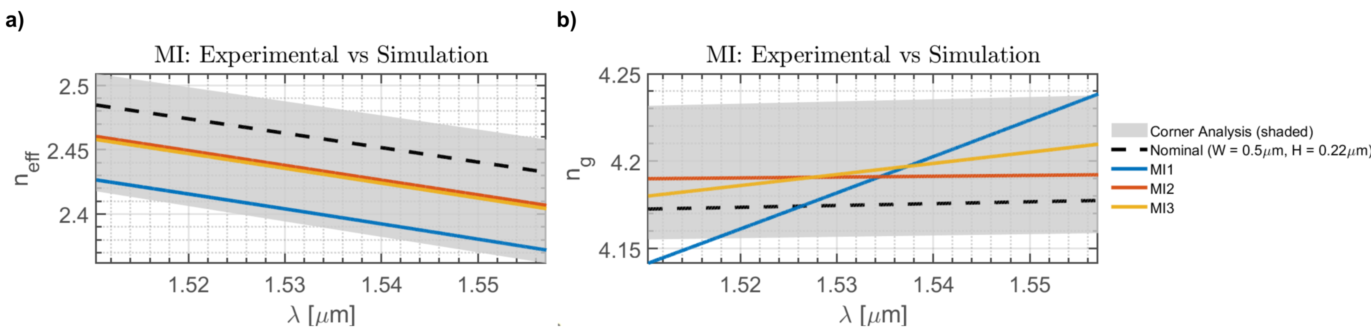

Fig. 20: Extracted (a) $\mathrm{n_{eff}}$ and (b) $\mathrm{n_g}$ from MIs. Colored curves show experimental results for MI1 – MZI3 with $\Delta L = 25, 50, 75$ μm. The black dashed line corresponds to the nominal waveguide geometry (W = 500 nm, H = 220 nm), while the shaded gray region represents the corner-analysis bounds.

Finally, Fig. 21 and 22 compares the experimentally measured FSR with the simulated FSR for all MZIs and MIs, respectively, where the simulations correspond only to the nominal geometry. The experimental results show excellent agreement with the simulations. This consistency further confirms the conclusions of the previous section, namely that fabrication-induced geometric variations have a negligible impact on the FSR, and therefore the MZI performance is highly robust, in terms of the FSR, to typical fabrication tolerances.

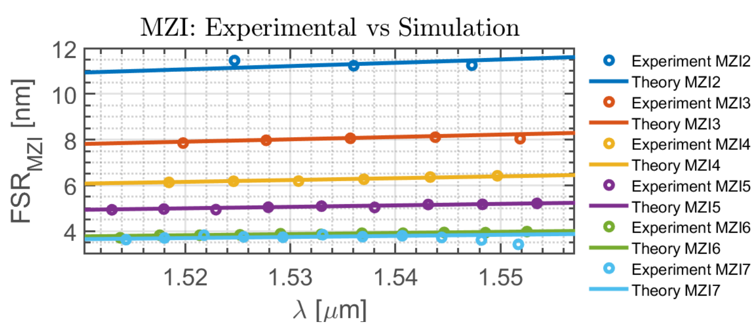

Fig. 21: Simulated (solid lines) versus Extracted experimental FSR (circular markers) for all the MZIs. The simulated results are for the nominal geometry of the waveguide.

VII. Conclusion

In this work, we have designed, fabricated, and characterized silicon-on-insulator Mach–Zehnder and Michelson interferometers for determining the effective and group indices of strip waveguides. The MZI comprises grating couplers, Y-branch splitters, and unbalanced optical paths, while the MI incorporates grating couplers, 3-dB couplers, mirror waveguides, and unbalanced arms. Experimental measurements enabled the extraction of both $n_{\rm eff}$ and $n_{\rm g}$, as well as the FSR for all fabricated MZIs and MIs, showing excellent agreement with design expectations with maximum deviations of only $\approx$1.5%. Corner analysis was also performed to account for fabrication tolerances, revealing the sensitivity of the effective and group indices to waveguide

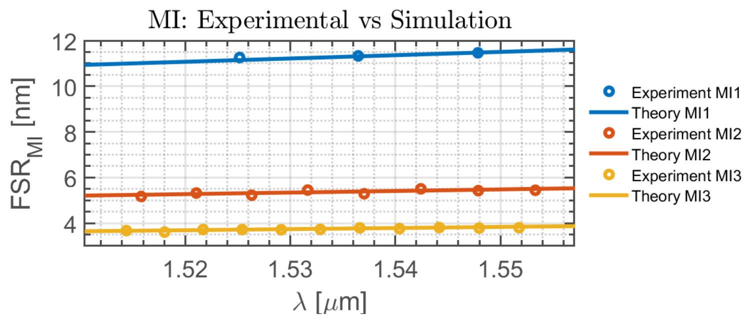

Fig. 22: Simulated (solid lines) versus Extracted experimental FSR (circular markers) for all the MIs. The simulated results are for the nominal geometry of the waveguide.

dimensional variations. The MI design demonstrated advantages in compactness and phase sensitivity compared to the MZI, confirming its suitability for accurate waveguide characterization. Overall, these results provide a comprehensive experimental validation of interferometric techniques for SOI waveguide index measurement.

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